

Soumik Kumar Basu

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EDUCATION

- **IIT Hyderabad** Kandi, Telengana
PhD in Computer Science and Engineering; CGPA: 9/10 Aug 2021 - Present
- **West Bengal University of Technology** Kolkata, West Bengal
Bachelor of Technology in Computer Science and Engineering; CGPA: 9.2/10 Aug 2017 - July 2021
- **Sinthi Ramkrishna Sangha Vidyamandir** Kolkata, West Bengal
Higher Secondary Examination; Percentage: 85.8% Aug 2015 - July 2017
- **Sinthi Ramkrishna Sangha Vidyamandir** Kolkata, West Bengal
Secondary Examination; Percentage: 91.9% Aug 2014 - July 2015

RESEARCH INTERESTS

Compilers, Program Analysis and Optimization, High-Performance Computing, Heterogeneous Computing

PUBLICATIONS

- **Soumik Kr. Basu**, Jyothi Vedurada, GSOHC: Global Synchronization Optimization for Heterogeneous Computing, ECOOP '25 (CORE Rank A)
- **Soumik Kr. Basu**, and Jyothi Vedurada, StreamAlloc: A Framework for Analyzing and Transforming CUDA Code to Enable Asynchronous Execution, TOPLAS (CORE Rank: A*) [*will also be presented in PLDI' 26*]
- **Soumik Kr. Basu**, and Jyothi Vedurada, UVMemcpy: A Framework for Optimizing Heterogeneous Programs That Use Unified Memory (Under Submission)

PATENTS

- **Soumik Kr. Basu**, Jyothi Vedurada, Method and System for Automatic CUDA Stream Allocation using Static Analysis, *Patent(IN) Granted in 2025*
- **Soumik Kr. Basu**, Jyothi Vedurada, Method and System for Automatically Optimizing Heterogeneous Programs That Use Unified Memory, *Patent(IN) Submitted in 2026*

RESEARCH FOCUS

My PhD work encompasses three distinct projects that primarily focus on compiler optimization opportunities on the host side of heterogeneous CPU-GPU programs to reduce execution time while minimizing manual programming effort.

HONORS AND AWARDS

- Winner of Distinguished Artifact Award in ECOOP 2025 for my paper, GSOHC
- Winner of Best Presentation Award in Indosys 2025 for presentation on my research paper, GSOHC
- Winner of Mathmania '18, an inter-college mathematics competition (Winner among 180 teams)
- Won Intra-School Science Acharya Medal for highest marks in Mathematics in Higher Secondary Exam
- Received "Certificate of Merit-2015" for "Excellent" performance in science subjects (99%) in Secondary Exam
- Received "Certificate of Merit-2017" for "Excellent" performance in Higher Secondary Exam

INVITED TALKS

- **Title:** GSOHC: Global Compiler Optimization for Heterogeneous Computing
 - **Venue:** Software Engineering Research in India (SERI), Warangal, 2025
 - **Venue:** Indosys, Bangalore, 2025
 - **Venue:** Innovation In Compiler Technology (IICT), Bangalore, 2025
 - **Venue:** ACM ARCS, Hyderabad, 2026
- **Title:** An Introduction of DSLs, **Venue:** ACM Summer School on Compilers for AI/ML, Pune, 2024
- **Title:** An Introduction to C++(Basic to Advanced)
Venue: ACM Summer School on Compilers for AI/ML, Pune, 2024
- **Title:** Understanding GPU Architectures and Programming
Venue: ACM IIT Hyderabad Student Chapter, 2023
- **Title:** Recent Optimization in GPU Architecture and Compilers
Venue: IIT Hyderabad CSE Seminar, 2023

SERVICES

- Served as a Program Committee [member](#) for PPOPP '23 (CORE Rank A) Artifact Evaluation Committee
- Served as a Moderator for Computer Systems in ACM IIT Hyderabad Student Chapter
- Served as a member of the Organizing Committee for CSI Regional Student Convention for Region - II '19
- Served as a Teaching Assistant for ACM Summer School on Compilers for AI / ML Programs, 2024

TEACHING EXPERIENCE

- **Computer Architecture(CS2323):** Teaching Assistant to Dr. Praveen Tammana (October '21 - Dec '21)
- **Operating Systems(CS3523):** Teaching Assistant to Dr. Sathya Peri (Jan '22 - Jun '22)
- **Software Engineering(CS4443):** Teaching Assistant to Dr. M.V. Panduranga Rao and Dr. Jyothi Vedurada (Jan '23 - Jun '23)
- **Compilers II(CS3423):** Teaching Assistant to Dr. Jyothi Vedurada (Aug '23 - Dec '23)
- **Introduction to Program Analysis and Optimization (CS5863):** Teaching Assistant to Dr. Jyothi Vedurada (Jan '24 - Jun '24), (Jan '25 - Jun '25), (Jan '26 - Jun '26)
- **Introduction to Programming (ID1063):** Teaching Assistant to Dr. Jyothi Vedurada, Dr. J. Saketha Nath, and Dr. Saurabh Kumar (Aug '24 - Dec '24)
- **GPU Programming (CS5013):** Teaching Assistant to Dr. Jyothi Vedurada (Aug '25 - Dec '25)

RELEVANT ACADEMIC COURSE WORKS

- **Introduction to Program Analysis and Optimization, IIT Hyderabad, Grade - 10/10 (A), Instructor: Dr. Jyothi Vedurada, Dr. Ramakrishna Upadrasta:** (Jan '22 - Jul '22)
- **Parallel and Concurrent Programming, IIT Hyderabad, Grade - 9/10 (A-), Instructor: Dr. Sathya Peri:** (Jan '22 - Jul '22)
- **Parallel Computing, IIT Hyderabad, Grade - 9/10 (A-), Instructor: Prof. C Siva Ram Murthy:** (Aug '22 - Nov '22)
- **Topics in Compiler Optimization, IIT Hyderabad, Grade - 10/10 (A), Instructor: Dr. Jyothi Vedurada and Dr. Ramakrishna Upadrasta:** (Aug '22 - Nov '22)

SKILLS

Languages/ APIs Known: CUDA, OpenCL, Java, C++, Python, C, SQL, Unix scripting, POSIX Threads,
Technical Tools: LLVM, Clang, Postgres, GDB, NVIDIA Nsight, **IDEs:** Visual Studio Code, Atom, Vim,
Other Tools: Latex, Git, Gnuplot, DOT, Docker

ACADEMIC PROJECTS

- **Concurrency Control in Transactional Systems, IIT Hyderabad, Instructor: Dr. Sathya Peri:** Topics Learned: Page and Object Models, Conflict Serializability, Linearizability, Concurrency Control. (Aug '21 - Dec '21) **Projects:** (1) Implemented two algorithms, BOCC-TS and ROMV-BOCC-TS from scratch using CPP Multithreading.
- **Parallel and Concurrent Programming, IIT Hyderabad, Instructor: Dr. Sathya Peri:** Topics Learned: Multi-Threading, Locking Protocols, Multi-Processing, Lock-free Systems, Mutex, Synchronization, Barrier Synchronization, MIMD programming paradigm, Parallel Algorithms (Jan '22 - Jul '22) **Projects:** (1) Developed a novel Parallel Kruskal's Algorithm using Naive Parallel (Static Thread Mapping) and Smart Parallel (Dynamic Thread Mapping) Approach.
- **Introduction to Program Analysis and Optimization, IIT Hyderabad, Instructor: Dr. Jyothi Vedurada:** Topics Learned: Killdal's Algorithm, Data Flow Analysis, Points-to Analysis, Escape Analysis, Compositional Bottom-up Analysis, Polyhedral and Loop Optimizations (Jan '22 - Jul '22) **Framework Developed:** (1) Developed an intra-procedural flow-sensitive top-down may alias analysis for C++ programs using LLVM/Clang infrastructure.

ONLINE COURSE WORKS

- **GPU Architectures and Programming:**
NPTEL, Instructor - Dr. Soumyajit De, IIT Kgp. (Jan' 22 - Apr' 22)¹
- **Fundamentals of Accelerated Computing with CUDA C/C++ :**
NVIDIA DLI Workshop. (Jun' 22)
- **Programming in JAVA:**
NPTEL, Instructor - Dr. Debasis Samanta, IIT Kgp. (Jan' 19 - Apr' 19)
- **Programming, Data Structures and Algorithms using Python:**
NPTEL, Instructor - Dr. Madhavan Mukund, IIT M. (Aug' 18 - Sep' 18)
- **Problem Solving through Programming in C:**
NPTEL, Instructor - Prof. Anupam Basu, IIT Kgp. (Jan' 18 - Apr' 18)²

¹Awarded ELITE with Silver certificate

²Awarded ELITE certificate